

Vedant Gupta

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EDUCATION

- **Dr. B.R. Ambedkar National Institute of Technology (NIT), Jalandhar** Jalandhar, Punjab
B.Tech in Electrical Engineering; CGPA: 8.50 Aug. 2024 – May 2028
- **City Montessori School, Lucknow** Lucknow, India
CISCE Board; Class XII: 96.5%, Class X: 97.6% June 2023
 - **Leadership:** Served as **Technical Prefect** at the world's largest school, spearheading **event management** and technical operations for city-wide assemblies and international competitions.

PROJECTS

- **StudyNotion — Scalable Ed-Tech Platform** [GitHub](#)
MERN Stack — React, Node.js, Express.js, MongoDB, Razorpay, Zustand, Tailwind CSS
 - **Architecture:** Built a full-stack MERN platform with **25+ REST APIs**, **15+ React pages**, and **10+ MongoDB models**, implementing **RBAC** for Student, Instructor, and Admin roles.
 - **Frontend:** Developed a **100% responsive UI** with **20+ reusable React components** and optimized global state management using **Zustand**.
 - **Security & Payments:** Implemented secure authentication using **JWT**, **Bcrypt**, and **OTP verification**, along with **Razorpay** integration for seamless course purchases.
 - **Media:** Integrated **Cloudinary** for scalable storage and delivery of course images and videos, ensuring efficient media management.
- **SmartNotes — AI-Powered Note-Taking Platform** [GitHub](#)
React.js, FastAPI, LangChain, Groq (LLaMA), FAISS, Faster Whisper, MongoDB, Firebase, Tailwind CSS
 - **AI Pipeline:** Built a full-stack AI platform supporting **4+ content formats** (YouTube, PDFs, MP3, and MP4) with automated summarization and conversational Q&A using **LangChain** and the **Groq Inference Engine**.
 - **RAG System:** Implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS**, generating embeddings for **2000+ document chunks** to enable source-grounded semantic retrieval.
 - **Backend & Processing:** Developed **15+ RESTful APIs** with **FastAPI** and integrated **Faster Whisper** to process and transcribe **2-hour videos** while delivering AI-generated responses in **under 2 seconds**.
 - **Performance Optimization:** Optimized the AI pipeline by parallel processing reducing document preparation time by **70%** and enabling faster end-to-end summarization.

SKILLS

- **Programming Languages:** C++, C, Python, JavaScript
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Socket.IO, NumPy, Pandas, Matplotlib, Scikit-Learn, Langchain, Pydantic, FastAPI
- **Databases & Tools:** MongoDB, Redis, Git, GitHub, Figma, Postman, PostgreSQL

ACHIEVEMENTS

- **Data Structures and Algorithms:** Solved **500+ problems** across various platforms, building a strong foundation in core concepts including Trees, Graphs, and Dynamic Programming.
- **Mathematics Olympiad:** Awarded **Bronze Medal** in the International Myanmar Mathematics Olympiad, demonstrating advanced analytical and problem-solving abilities.
- **JEE Mains:** Secured **98.58 percentile**, ranking in the top tier of candidates nationally in one of the world's most competitive engineering entrance examinations.